

Reviewer Pack — Coxeter-Dynkin Diagrams and Circuit Depth Inequality

Entry f588f750e146 · INCONCLUSIVE

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Coxeter-Dynkin Diagrams and Circuit Depth Inequality

Entry ID: f588f750e146 Verdict: INCONCLUSIVE Recorded: 2026-06-05 15:46:12 UTC

1. Conjecture

Field A (mathematical branch): Combinatorial Geometry (Coxeter-Dynkin Diagrams) **Field B** (complexity object): Boolean Circuit Complexity (Circuit Depth)

Statement:

For every boolean circuit C with depth d , the number of nodes in its associated Coxeter-Dynkin diagram is at most a polynomial in d .

Rationale (proposer’s reasoning):

The Coxeter-Dynkin diagrams encode symmetries and automorphisms of polyhedra. Since the depth of a circuit relates to the number of levels of computation, a relationship between these two might reveal a geometric interpretation of computational complexity.

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 05582867bb22cc57

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if, for all circuits C with depth $d \leq 10$, the number of nodes in its Coxeter-Dynkin diagram is less than or equal to a polynomial function of d with a degree no greater than 3. The criterion is falsified if there exists a circuit with depth $d \leq 10$ such that the number of nodes in its Coxeter-Dynkin diagram exceeds a polynomial function of d with a degree no greater than 3.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	HITS	SAFE
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - (Coxeter-Dynkin Diagrams) AND (circuit depth) AND (polynomial inequality) - (Boolean circuit complexity) AND (depth) AND (Coxeter diagrams) - (Coxeter-Dynkin diagrams) IN title AND (circuit depth) IN title AND polynomial

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=0, elapsed=0.1s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_circuit(depth):
        if depth == 0:
            return ['input']
        else:
            left = generate_circuit(depth - 1)
            right = generate_circuit(depth - 1)
            return [f'AND({left[0]}, {right[0]})', f'OR({left[0]}, {right[0]})']

    def count_nodes(circuit):
        if isinstance(circuit, str):
            return 1
        else:
            return 1 + sum(count_nodes(sub_circuit) for sub_circuit in circuit)

    depths = [5, 10]
    total_nodes = 0
    instances_tested = 0
    n_max = 0

    for depth in depths:
        for _ in range(3):
            circuit = generate_circuit(depth)
            nodes = count_nodes(circuit)
            if nodes > n_max:
                n_max = nodes
            total_nodes += nodes
            instances_tested += 1

    conjecture_holds = all(nodes <= 2 * depth**3 for depth in depths for _ in range(3))
    counterexample = "Depth {}, Nodes {}".format(depths[0], total_nodes) if not conjecture_holds else ""
```


Critic reasoning:

The test only considers a small range of depths (5 and 10) and only three instances per depth, which is insufficient to confirm the conjecture for all boolean circuits. The metric does not scale trivially with n , but the sample size is too small to draw a robust conclusion.

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test only considers a small range of depths (5 and 10) with limited instances per depth, which is insufficient to confirm the conjecture for all boolean circuits. The critic challenges the robustness of the conclusion due to the small sample size. | next: Increase the number of tested circuits across a wider range of depths to provide a more comprehensive validation of the conjecture.

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	14011
2	preregistration	ollama_remote	glm4:latest	0	0	9829
3	novelty	ollama_remote	glm4:latest	0	0	8472
4	novelty	ollama_remote	glm4:latest	0	0	9300
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12480
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9295
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7585
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7998
9	critic	ollama_remote	glm4:latest	0	0	21620
10	judge	ollama_remote	glm4:latest	0	0	9169

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 109760 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in `research/audit/f588f750e146.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/f588f750e146.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/f588f750e146.tar.gz` (if generated)